State Machines – Part 6
K. Sridharan

Review

- We have given a design for a sequence detector using D flip-flops.
- We will now design a 3-bit binary counter using T flip-flops.

How is design with T flip-flops different ?

- We have designed a sequence detector with D flip-flops.
- We indicated that the relationship between next state and input D (for a D flip-flop) is direct.
- In the case of JK and T flip-flops, the relationship is not direct. As a result, we need a mechanism to connect the state table and input equations.

Use of an excitation table

- During the design process, we know from what state we are moving to another state but we need to find out the flip-flop input conditions that will cause the required transition.
- Using the characteristic tables described earlier for T flip-flop (see part 3 in sequential logic circuits),

  we can develop a table that lists the required inputs for a specific state change. This table is called the excitation table. We will recall the characteristic tables for the flip-flops and then present the excitation tables.

Characteristic Tables

JK Flip-Flop

J	K	Q(t + 1)	
0	0	$Q(t)$	No change
0	1	0	Reset
1	0	1	Set
1	1	$Q'(t)$	Complement

D Flip-Flop

D	Q(t + 1)	
0	0	Reset
1	1	Set

T Flip-Flop

T	Q(t + 1)	
0	$Q(t)$	No change
1	$Q'(t)$	Complement

On the characteristic tables

- We can observe from the characteristic tables that the relationship between next state and flip-flop input is direct only for a D flip-flop.
- We will use the characteristic tables for the other two flip-flops (namely, JK and T) to draw the excitation table.
- The excitation tables are shown next.

Excitation Tables

Q(t)	Q(t = 1)	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

(a) *JK* Flip-Flop

Q(t)	Q(t = 1)	T
0	0	0
0	1	1
1	0	1
1	1	0

(b) *T* Flip-Flop

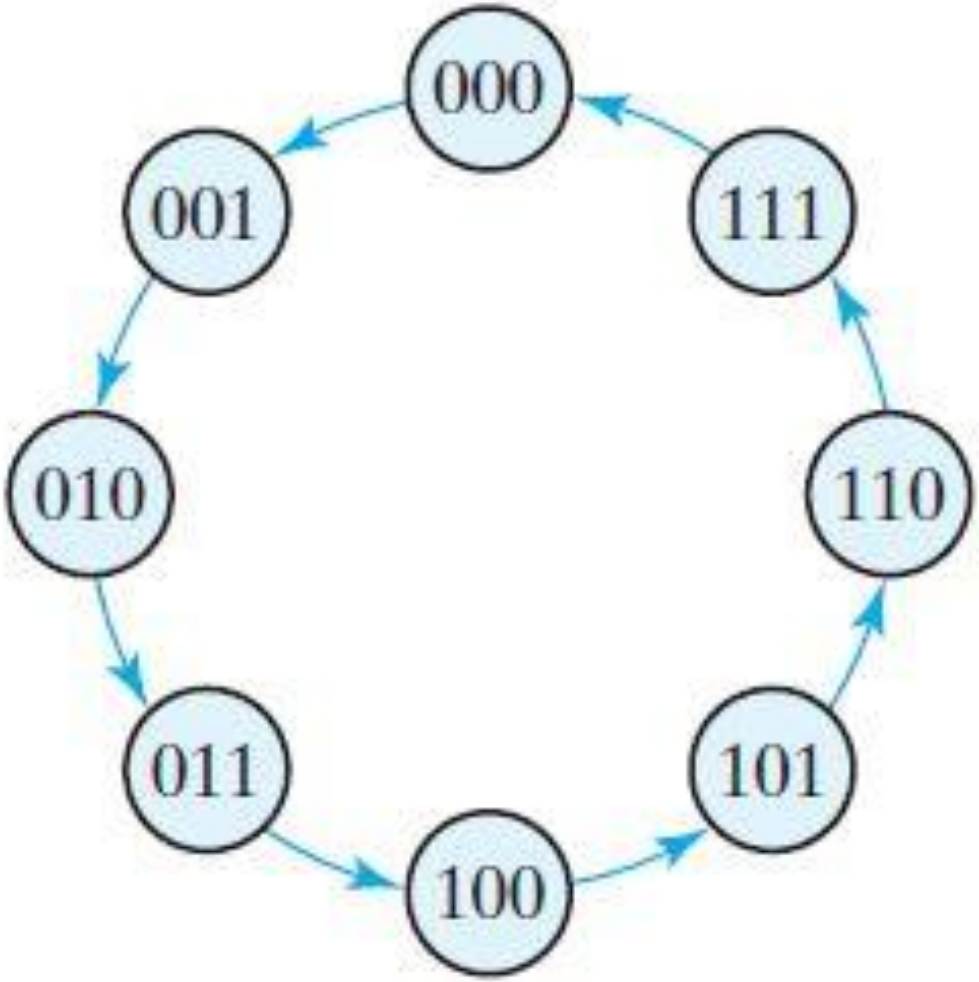
Explanation

- From the characteristic table of a T flip-flop, we observe that when input T = 1, the state of the flip-flop is complemented. When T = 0, the flip-flop's state remains unchanged.
- We can use this information to "move in the other direction", namely from the state to the input: when the state of the flip-flop should remain the same, the requirement is T = 0. When the state of the flip-flop has to be complemented, T must equal 1. This is what the excitation table conveys. We use this to design the 3-bit counter.

Design of a 3-bit binary counter

- The counter is intended to count from 0 to 7 and then repeat the same sequence.
- A counter of this kind does not require any input other than a clock.
- As indicated earlier, a clock does not explicitly appear in state diagrams or state tables. The state diagram of the counter is shown next.

State Diagram



Explanation for the state diagram

- The diagram does not have to show input and output values along the directed lines.
- The only input to the circuit is the clock while the outputs are specified by the present state of the flip-flops.
- The next state of the counter depends entirely on its present state and state transition is decided by the clock signal.

State Table

Present State			Next State			Flip-Flop Inputs		
A_2	A_1	A_0	A_2	A_1	A_0	T_{A2}	T_{A1}	T_{A0}
0	0	0	0	0	1	0	0	1
0	0	1	0	1	0	0	1	1
0	1	0	0	1	1	0	0	1
0	1	1	1	0	0	1	1	1
1	0	0	1	0	1	0	0	1
1	0	1	1	1	0	0	1	1
1	1	0	1	1	1	0	0	1
1	1	1	0	0	0	1	1	1

Explanation

- There are three flip-flops denoted by the symbols A_2 , A_1 and A_0 .
- We have already seen the excitation table for the T flip-flop. We know from the excitation table the input for T that is responsible for a specific state change.
- For example, consider row 1 in the state table: A_2 does not undergo any change so T_{A2} input should have been 0. Similarly, A_0 changes from 0 to 1 so T_{A0} should have been 1.

More on the state table

- Similarly, consider row 4: we go from 0 1 1 to 1 0 0. So all the three flip-flop outputs are changing. This means the toggle inputs for all the three flip-flops should have been 1.
- This way, one can explain the entries in all the rows pertaining to the flip-flop inputs.
- Once we have all the flip-flop inputs, the next step is to express T_{A2} , T_{A1} and T_{A0} in terms of A_2, A_1 and A_0

Simplification via Karnaugh maps

- The simplified form of expressions for T_{A2} , T_{A1} and T_{A0} can be obtained via Karnaugh maps. In particular,

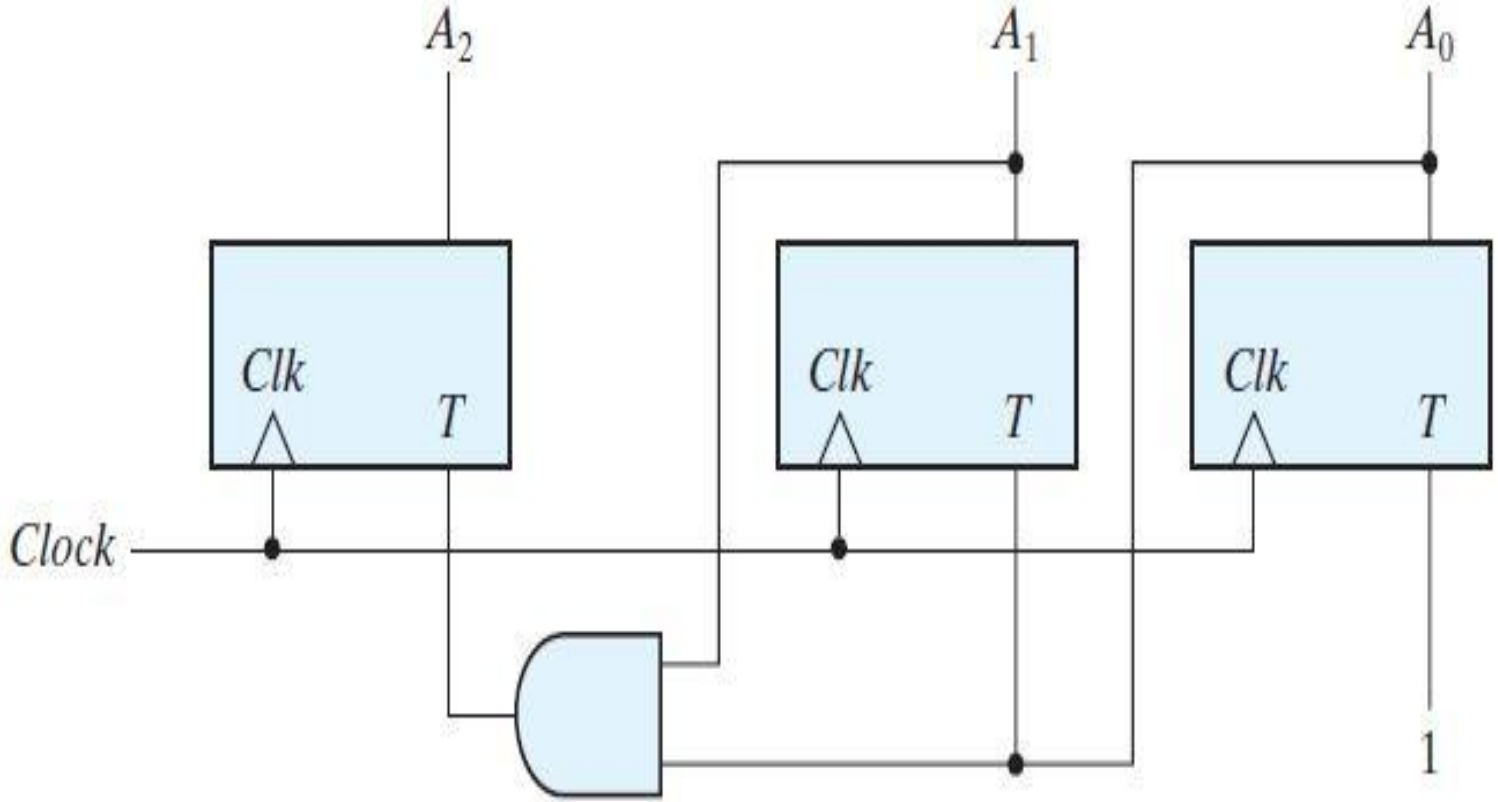
$$T_{A2} = A_1 A_0$$

$$T_{A1} = A_0$$

$$T_{A0} = 1$$

So we can see that for the `rightmost' flip-flop, there is a toggle every time leading to $T_{A0} = 1$

The circuit



Recall …

- We have shown a circuit similar to the one in the previous slide in Topic 6 (see part 5 in Sequential Logic).
- There we had a 4-bit up counter (and consequently one more T flip-flop and one more AND gate).
- What we have done now is to give a formal design procedure.

Summary

- We are done with state machines.
- We have discussed analysis as well as design of sequential circuits.

Textbook and Reference

- M. Mano and M. Ciletti, Digital Design, Pearson Education, 2018
- C. Roth and L.L. Kinney, Fundamentals of Logic Design, Gamma Publications, 2021.